

4-D Power is similar to 1-D w/ space (x, y, z)

S = complex (VA)

P = real Power (Watts) [Radiated] } optics

Q = imaginary Power (VARs) [Evanescent] }

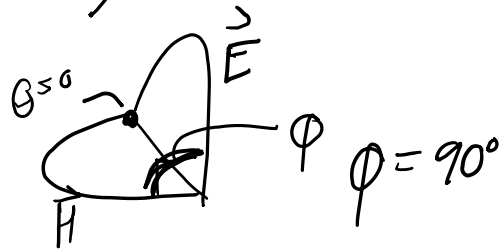
* Freespace (Assumption)

$$\vec{S} = \vec{E} \times \vec{H} = |\vec{E}| |\vec{H}| \sin \phi$$

$$\vec{E} = E_0 \sin(\omega t - kz) [\vec{u}_e]$$

$$\vec{H} = H_0 \sin(\omega t - kz + \theta) [\vec{u}_h] \quad \theta \text{ is the time angle b/w } E \text{ and } H$$

$$B = \mu \vec{H}$$



$$S(\text{Poynting vector}) = E_0 \sin(\omega t - kz) H_0 \sin(\omega t - kz + \theta) [\vec{u}_k] \quad \theta \text{ is timing difference.}$$

$$\frac{E_0}{H_0} = 377 \Omega$$

$$H_0 = \frac{E_0}{377}$$

ϕ = is spatial angle b/w $\vec{E}$ and $\vec{H}$

θ = is temporal angle b/w $\vec{E}$ and $\vec{H}$

Φ = is spatial angle difference b/w two locations on the same wave.

$$\vec{S} = E_0 \sin(\omega t - kz) \frac{E_0}{377} \sin(\omega t - kz + \theta)$$

$$a = \omega t - kz$$

$$b = \theta$$

$$E = E_0 \sin(a)$$

$$H = \frac{E_0}{377} \sin(a+b)$$

$$S = \frac{E_0^2}{377} \left(\sin(a) \sin(a+b) \right) \rightarrow \text{Trig I.D.} \quad \sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$

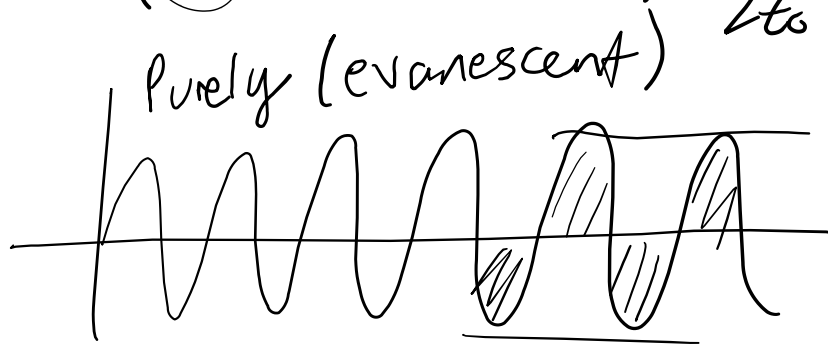
$$= \frac{E_0^2}{377} \left(\sin(a) \left[\sin(a)\cos(b) + \cos(a)\sin(b) \right] \right)$$

$$= \frac{E_0^2}{377} \left(\underbrace{\sin^2(a)} \cos(b) + \underbrace{\sin(a)\cos(a)} \sin(b) \right)$$

Trig I.D.

$$= \frac{E_0^2}{377} \left(\left[\frac{1 - \cos(2a)}{2} \right] \cos(b) + \frac{1}{2} \sin(2a) \sin(b) \right)$$

$$\vec{S} = \frac{E_0^2}{2(Z_0)} \cos \Theta \left(\underbrace{1}_{\text{DC offset}} - \cos(2(\omega t - kz)) \right) + \frac{E_0^2}{2Z_0} \sin \Theta \left(\sin(2(\omega t - kz)) \right)$$



Mix of $P + jQ$



$$S_{ave} = \frac{E_0^2}{2(Z_0)} \cos \theta \quad \left[\frac{W}{m^2} \right]$$

S_{ave} is called Irradiance

Irradiance refers to what amount of energy per time (Power) that arrives on a certain surface with a given orientation